Compound interest tables

1 $\$4400$ is invested for three years at a rate of 5% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	–	$\$4400$	$\$220$
Second year	$\$4400 + \$220$	$\$4620$	$\$231$
Third year	$\$4620 + \$▯$	$\$▯$	$\$▯$
Fourth year	$\$4851 + \$▯$	$\$▯$	–

b Calculate the total interest earned over the three years.

2 $\$3600$ is invested for three years at a rate of 7% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	–	$\$3600$	$\$252$
Second year	$\$3600 + \$252$	$\$3852$	$\$269.64$
Third year	$\$3852 + \$▯$	$\$▯$	$\$▯$
Fourth year	$\$4121.64 + \$▯$	$\$▯$	–

b Calculate the total interest earned over the three years.

3 $\$570$ is invested for two years at a rate of 9% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	–	$\$570$	$\$▯$
Second year	$\$570 + \$▯$	$\$▯$	$\$▯$
Third year	$\$▯ + \$55.92$	$\$▯$	–

b Calculate the total interest earned over the three years.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 $\$6100$ is invested for two years at a rate of 6% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	–	$\$6100$	$\$$▯
Second year	$\$6100 + \$$▯	$\$$▯	$\$$▯
Third year	$\$$▯ $+ \$387.96$	$\$$▯	–

b Calculate the total interest earned over the three years.

5 $\$1000$ is invested at 2% per year, compounded annually. The table tracks the growth of the principal over three years.

Time Period n (years)	Value at beginning of time period	Value at end of time period	Interest earned in time period
1	$\$1000$	$\$$▯	$\$$▯
2	$\$$▯	$\$1040.40$	$\$$▯
3	$\$1040.40$	$\$1061.21$	$\$$▯

a Copy and complete the table.

b Calculate the total interest earned over the three years.

Compound interest

6 State whether the following statements are true or false about compound interest:

a Interest is earned on the principal.

b The interest in any time period is calculated using only the original principal.

c Interest is earned on any accumulated interest.

d The amount of interest earned in any time period changes from one period to the next.

7 Roberto's investment of $\$70\,000$ earns interest at 5% per year, compounded annually over 3 years.

a Calculate the value of the investment after 3 years.

b Calculate the amount of interest earned.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Maria invested $\$8090$ at 8% per year, compounded annually over 3 years.
Answer the following questions without using the compound interest formula.

- **a** What was the interest for the first year?
- **b** What was the total amount invested after the first year?
- **c** What was the interest for the second year?
- **d** What was the total amount invested after the second year?
- **e** What was the interest for the third year?
- **f** What was the total amount invested after the third year?
- **g** What was the total amount of interest earned over the 3 years?
- **h** What was the interest as a percentage of the initial investment? Give your answer as a percentage to two decimal places.
- **i** What would have been the simple interest earned after 3 years?
- **j** How much more compound interest than simple interest would have accumulated over these 3 years?

9 Hannah borrows $\$5000$ at a rate of 4.2% per year, compounded annually.

- **a** After 4 years, Hannah repays the loan all at once. How much money does she pay back in total?
- **b** How much interest was generated on the loan over the four years?

10 Jerome has just won $\$50\,000$. When he retires in 15 years, he wants to have $\$94\,000$ in his fund which earns 6% interest annually. Determine how much of his winnings he needs to invest now to achieve this.

11 Victoria has been promised an inheritance of $\$70\,000$ in 5 years time. Determine the maximum she can borrow now at a rate of 7% compounded annually, and still be able to pay off the loan with her inheritance.

12 When Yuri got his first job, he invested $\$10\,000$ in an account with a 1.3% annual growth rate.

- **a** Write a function, $A(t)$, that represents the value of this investment t years after Yuri started his job.
- **b** Calculate how much more the investment will be worth when Yuri has been working for 12 years than when he had been working for 5 years.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Other compound periods

13 Calculate the value of the following investments:

a A $\$7230$ investment earns interest at 3% per year, compounded quarterly over 7 years.

b A $\$5520$ investment earns interest at 0.8% per year, compounded monthly over 2 years.

c A $\$1980$ investment earns interest at 0.45% per year, compounded weekly over 2 years.

d A $\$3000$ investment earns interest at 4.2% per year, compounded daily over 14 years.

14 Calculate the amount of interest earned by the following investments:

a Beth's investment of $\$4800$ earns interest at 3.3% per year, compounded monthly over 13 years.

b James's investment of $\$4210$ earns interest at 0.78% per year, compounded weekly over 16 years.

c Davinda's investment of $\$5390$ earns interest at 2.1% per year, compounded daily over 11 years.

d Una's investment of $\$80\,000$ earns interest at 0.5% per year, compounded semi-annually over 10 years.

15 Amelia's investment into a 15-year 4.09% per year. corporate bond grew to $\$14\,160$. Calculate the size of Amelia's initial investment if interest was compounded:

a Annually **b** Semiannually **c** Quarterly **d** Monthly

e Weekly **f** Daily

16 Grace wants to invest $\$1600$ at 5% per year for 7 years. She has three investment options, compounded quarterly, compounded monthly, or compounded weekly.

a Calculate the value of the investment if it is compounded:

i quarterly **ii** monthly **iii** weekly

b Find the difference between the final investment values of the three options.

c Describe the effect that changing the compound frequency has on the interest earned.

17 Maria has $\$5000$ to invest for 6 years and would like to know which investment plan to enter into out of the following three:

- Plan 1: invest at 4.70% per year interest, compounded monthly
- Plan 2: invest at 5.77% per year interest, compounded quarterly
- Plan 3: invest at 5.40% per year interest, compounded annually

Determine which is the best option for Maria. Use calculations and reasoning to explain your

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

© ClipSAT: By Mr. Mohamed Abdallah +966597688647